Laboratorio 7 - Visión por Computadora

- Paula Barillas - 22764
- Gerardo Pineda - 22880
- Mónica Salvatierra - 22249

Link del repositorio: https://github.com/alee2602/LAB7-VCP

Task 3

Imports

In [60]:
```
import os
from pathlib import Path
import pandas as pd
from sklearn.model_selection import train_test_split
import torch
from torch.utils.data import Dataset, DataLoader
from PIL import Image
from torchvision import transforms
import matplotlib.pyplot as plt
from torchvision import models
import torch.nn as nn
import copy
import time
import numpy as np
from sklearn.metrics import accuracy_score, f1_score, recall_score
```

Definir ruta del dataset

In [2]:
```
DATASET_DIR = Path("data/chest_xray")

print(DATASET_DIR.exists())
print(list(DATASET_DIR.iterdir()))
```

```
True
[WindowsPath('data/chest_xray/test'), WindowsPath('data/chest_xray/train'), WindowsP
ath('data/chest_xray/val')]
```

Obtener rutas y etiquetarlas

In [3]:
```
def collect_paths(dataset_dir):
    records = []
    splits = ["train", "val", "test"]
    classes = ["NORMAL", "PNEUMONIA"]
```

```
    for split in splits:
        for class_name in classes:
            class_dir = dataset_dir / split / class_name

            if not class_dir.exists():
                print(f"Advertencia: no existe {class_dir}")
                continue

            for img_path in class_dir.glob("*"):
                if img_path.is_file():
                    records.append({
                        "filepath": str(img_path),
                        "label_name": class_name,
                        "label": 0 if class_name == "NORMAL" else 1,
                        "original_split": split
                    })

    return pd.DataFrame(records)
```

Unificar datatset

In [4]:
```
df = collect_paths(DATASET_DIR)

print("Total de imágenes:", len(df))
df.head()
```

```
Total de imágenes: 5856
```

Out[4]:

	filepath	label_name	label	original_split
0	data\chest_xray\train\NORMAL\IM-0115-0001.jpeg	NORMAL	0	train
1	data\chest_xray\train\NORMAL\IM-0117-0001.jpeg	NORMAL	0	train
2	data\chest_xray\train\NORMAL\IM-0119-0001.jpeg	NORMAL	0	train
3	data\chest_xray\train\NORMAL\IM-0122-0001.jpeg	NORMAL	0	train
4	data\chest_xray\train\NORMAL\IM-0125-0001.jpeg	NORMAL	0	train

Obtener distribución general

In [5]:
```
print(df["label_name"].value_counts())
print()
print(df["label_name"].value_counts(normalize=True))
```

```
label_name
PNEUMONIA    4273
NORMAL       1583
Name: count, dtype: int64

label_name
PNEUMONIA    0.729679
NORMAL       0.270321
Name: proportion, dtype: float64
```

Observaciones: Las clases se encuentran desbalanceadas, puesto que existen más imágenes de pulmones con pneumonía bacteriana que sin

Realizar split estratificado

In [6]:
```
train_df, temp_df = train_test_split(
    df,
    test_size=0.30,
    stratify=df["label"],
    random_state=42
)

val_df, test_df = train_test_split(
    temp_df,
    test_size=0.50,
    stratify=temp_df["label"],
    random_state=42
)

print("Tamaño total:", len(df))
print("Train:", len(train_df), f"({len(train_df)/len(df):.2%})")
print("Val:", len(val_df), f"({len(val_df)/len(df):.2%})")
print("Test:", len(test_df), f"({len(test_df)/len(df):.2%})")
```

```
Tamaño total: 5856
Train: 4099 (70.00%)
Val: 878 (14.99%)
Test: 879 (15.01%)
```

In [8]:
```
def show_distribution(name, subset_df):
    print(f"--- {name} ---")
    print(subset_df["label_name"].value_counts())
    print()
    print(subset_df["label_name"].value_counts(normalize=True))
    print()

show_distribution("TRAIN", train_df)
show_distribution("VALIDATION", val_df)
show_distribution("TEST", test_df)
```

```
--- TRAIN ---
label_name
PNEUMONIA    2991
NORMAL       1108
Name: count, dtype: int64

label_name
PNEUMONIA    0.72969
NORMAL       0.27031
Name: proportion, dtype: float64

--- VALIDATION ---
label_name
PNEUMONIA    641
NORMAL       237
Name: count, dtype: int64

label_name
PNEUMONIA    0.730068
NORMAL       0.269932
Name: proportion, dtype: float64

--- TEST ---
label_name
PNEUMONIA    641
NORMAL       238
Name: count, dtype: int64

label_name
PNEUMONIA    0.729238
NORMAL       0.270762
Name: proportion, dtype: float64
```

Resetear índices

In [7]:
```
train_df = train_df.reset_index(drop=True)
val_df = val_df.reset_index(drop=True)
test_df = test_df.reset_index(drop=True)
```

In [8]:
```
train_df.to_csv("train_split.csv", index=False)
val_df.to_csv("val_split.csv", index=False)
test_df.to_csv("test_split.csv", index=False)

print("Splits guardados en CSV.")
```

```
Splits guardados en CSV.
```

In [9]:
```
IMAGE_SIZE = 224
BATCH_SIZE = 32
NUM_WORKERS = 0
```

Data Augmentation

In [16]:
```
train_transform = transforms.Compose([
    transforms.Resize((IMAGE_SIZE, IMAGE_SIZE)),
    transforms.RandomRotation(degrees=5),
    transforms.RandomAffine(
        degrees=0,
        translate=(0.02, 0.02),
        scale=(0.98, 1.02)
    ),
    transforms.ToTensor(),
    transforms.Normalize(
        mean=[0.485, 0.456, 0.406],
        std=[0.229, 0.224, 0.225]
    )
])

eval_transform = transforms.Compose([
    transforms.Resize((IMAGE_SIZE, IMAGE_SIZE)),
    transforms.ToTensor(),
    transforms.Normalize(
        mean=[0.485, 0.456, 0.406],
        std=[0.229, 0.224, 0.225]
    )
])
```

Datatset personalizado

In [17]:
```
class ChestXRayDataset(Dataset):
    def __init__(self, dataframe, transform=None):
        self.dataframe = dataframe
        self.transform = transform

    def __len__(self):
        return len(self.dataframe)

    def __getitem__(self, idx):
        row = self.dataframe.iloc[idx]
        img_path = row["filepath"]
        label = int(row["label"])

        image = Image.open(img_path).convert("RGB")

        if self.transform:
            image = self.transform(image)

        return image, label
```

In [18]:
```
train_dataset = ChestXRayDataset(train_df, transform=train_transform)
val_dataset = ChestXRayDataset(val_df, transform=eval_transform)
test_dataset = ChestXRayDataset(test_df, transform=eval_transform)
```

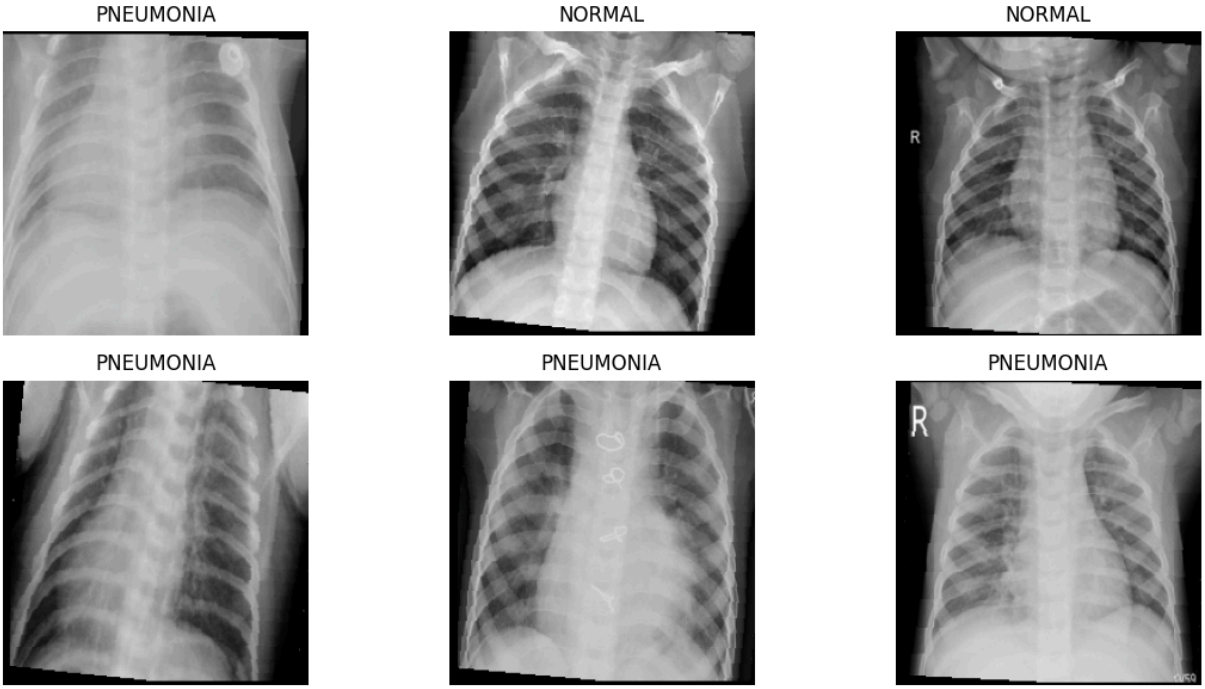
Crear dataloaders

In [19]:
```
train_loader = DataLoader(
    train_dataset,
    batch_size=BATCH_SIZE,
    shuffle=True,
    num_workers=NUM_WORKERS
)

val_loader = DataLoader(
    val_dataset,
    batch_size=BATCH_SIZE,
    shuffle=False,
    num_workers=NUM_WORKERS
)

test_loader = DataLoader(
    test_dataset,
    batch_size=BATCH_SIZE,
    shuffle=False,
    num_workers=NUM_WORKERS
)
```

In [20]:
```
images, labels = next(iter(train_loader))

print("Batch de imágenes:", images.shape)
print("Batch de labels:", labels.shape)
print("Labels ejemplo:", labels[:10])
```

```
Batch de imágenes: torch.Size([32, 3, 224, 224])
Batch de labels: torch.Size([32])
Labels ejemplo: tensor([1, 1, 1, 0, 1, 0, 1, 1, 1, 1])
```

In [ ]:
```
def imshow_tensor(img_tensor):
    img = img_tensor.numpy().transpose((1, 2, 0))
    mean = np.array([0.485, 0.456, 0.406])
    std = np.array([0.229, 0.224, 0.225])
    img = std * img + mean
    img = np.clip(img, 0, 1)
    plt.imshow(img)
    plt.axis("off")

images, labels = next(iter(train_loader))

plt.figure(figsize=(12, 6))
for i in range(6):
    plt.subplot(2, 3, i + 1)
    imshow_tensor(images[i])
    plt.title("PNEUMONIA" if labels[i].item() == 1 else "NORMAL")
plt.tight_layout()
plt.show()
```



In [ ]:
```
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print("Usando dispositivo:", device)
```

```
Usando dispositivo: cuda
```

Congelar capas y entrenamiento de modelos

In [22]:
```
def freeze_model_base(model):
    for param in model.parameters():
        param.requires_grad = False
```

In [25]:
```
mobilenet = models.mobilenet_v2(weights=models.MobileNet_V2_Weights.DEFAULT)

# congelar toda la base
freeze_model_base(mobilenet)

# reemplazar clasificador final
in_features_mobilenet = mobilenet.classifier[1].in_features

mobilenet.classifier = nn.Sequential(
    nn.Dropout(p=0.2),
    nn.Linear(in_features_mobilenet, 2)
)

mobilenet = mobilenet.to(device)
print(mobilenet)
```

```
MobileNetV2(
  (features): Sequential(
    (0): Conv2dNormActivation(
      (0): Conv2d(3, 32, kernel_size=(3, 3), stride=(2, 2), padding=(1, 1), bias=Fal
se)
      (1): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats
=True)
      (2): ReLU6(inplace=True)
    )
    (1): InvertedResidual(
      (conv): Sequential(
        (0): Conv2dNormActivation(
          (0): Conv2d(32, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), gro
ups=32, bias=False)
          (1): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_s
tats=True)
          (2): ReLU6(inplace=True)
        )
        (1): Conv2d(32, 16, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (2): BatchNorm2d(16, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      )
    )
    (2): InvertedResidual(
      (conv): Sequential(
        (0): Conv2dNormActivation(
          (0): Conv2d(16, 96, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(96, eps=1e-05, momentum=0.1, affine=True, track_running_s
tats=True)
          (2): ReLU6(inplace=True)
        )
        (1): Conv2dNormActivation(
          (0): Conv2d(96, 96, kernel_size=(3, 3), stride=(2, 2), padding=(1, 1), gro
ups=96, bias=False)
          (1): BatchNorm2d(96, eps=1e-05, momentum=0.1, affine=True, track_running_s
tats=True)
          (2): ReLU6(inplace=True)
        )
        (2): Conv2d(96, 24, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (3): BatchNorm2d(24, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      )
    )
    (3): InvertedResidual(
      (conv): Sequential(
        (0): Conv2dNormActivation(
          (0): Conv2d(24, 144, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(144, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
          (2): ReLU6(inplace=True)
        )
        (1): Conv2dNormActivation(
          (0): Conv2d(144, 144, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), g
roups=144, bias=False)
          (1): BatchNorm2d(144, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
```

```
          (2): ReLU6(inplace=True)
        )
        (2): Conv2d(144, 24, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (3): BatchNorm2d(24, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      )
    )
    (4): InvertedResidual(
      (conv): Sequential(
        (0): Conv2dNormActivation(
          (0): Conv2d(24, 144, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(144, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
          (2): ReLU6(inplace=True)
        )
        (1): Conv2dNormActivation(
          (0): Conv2d(144, 144, kernel_size=(3, 3), stride=(2, 2), padding=(1, 1), g
roups=144, bias=False)
          (1): BatchNorm2d(144, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
          (2): ReLU6(inplace=True)
        )
        (2): Conv2d(144, 32, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (3): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      )
    )
    (5): InvertedResidual(
      (conv): Sequential(
        (0): Conv2dNormActivation(
          (0): Conv2d(32, 192, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(192, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
          (2): ReLU6(inplace=True)
        )
        (1): Conv2dNormActivation(
          (0): Conv2d(192, 192, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), g
roups=192, bias=False)
          (1): BatchNorm2d(192, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
          (2): ReLU6(inplace=True)
        )
        (2): Conv2d(192, 32, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (3): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      )
    )
    (6): InvertedResidual(
      (conv): Sequential(
        (0): Conv2dNormActivation(
          (0): Conv2d(32, 192, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(192, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
          (2): ReLU6(inplace=True)
        )
        (1): Conv2dNormActivation(
```

```
          (0): Conv2d(192, 192, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), g
roups=192, bias=False)
          (1): BatchNorm2d(192, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
          (2): ReLU6(inplace=True)
        )
        (2): Conv2d(192, 32, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (3): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      )
    )
    (7): InvertedResidual(
      (conv): Sequential(
        (0): Conv2dNormActivation(
          (0): Conv2d(32, 192, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(192, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
          (2): ReLU6(inplace=True)
        )
        (1): Conv2dNormActivation(
          (0): Conv2d(192, 192, kernel_size=(3, 3), stride=(2, 2), padding=(1, 1), g
roups=192, bias=False)
          (1): BatchNorm2d(192, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
          (2): ReLU6(inplace=True)
        )
        (2): Conv2d(192, 64, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (3): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      )
    )
    (8): InvertedResidual(
      (conv): Sequential(
        (0): Conv2dNormActivation(
          (0): Conv2d(64, 384, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(384, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
          (2): ReLU6(inplace=True)
        )
        (1): Conv2dNormActivation(
          (0): Conv2d(384, 384, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), g
roups=384, bias=False)
          (1): BatchNorm2d(384, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
          (2): ReLU6(inplace=True)
        )
        (2): Conv2d(384, 64, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (3): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      )
    )
    (9): InvertedResidual(
      (conv): Sequential(
        (0): Conv2dNormActivation(
          (0): Conv2d(64, 384, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(384, eps=1e-05, momentum=0.1, affine=True, track_running_
```

```
stats=True)
          (2): ReLU6(inplace=True)
        )
        (1): Conv2dNormActivation(
          (0): Conv2d(384, 384, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), g
roups=384, bias=False)
          (1): BatchNorm2d(384, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
          (2): ReLU6(inplace=True)
        )
        (2): Conv2d(384, 64, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (3): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      )
    )
    (10): InvertedResidual(
      (conv): Sequential(
        (0): Conv2dNormActivation(
          (0): Conv2d(64, 384, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(384, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
          (2): ReLU6(inplace=True)
        )
        (1): Conv2dNormActivation(
          (0): Conv2d(384, 384, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), g
roups=384, bias=False)
          (1): BatchNorm2d(384, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
          (2): ReLU6(inplace=True)
        )
        (2): Conv2d(384, 64, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (3): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      )
    )
    (11): InvertedResidual(
      (conv): Sequential(
        (0): Conv2dNormActivation(
          (0): Conv2d(64, 384, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(384, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
          (2): ReLU6(inplace=True)
        )
        (1): Conv2dNormActivation(
          (0): Conv2d(384, 384, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), g
roups=384, bias=False)
          (1): BatchNorm2d(384, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
          (2): ReLU6(inplace=True)
        )
        (2): Conv2d(384, 96, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (3): BatchNorm2d(96, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      )
    )
    (12): InvertedResidual(
```

```
      (conv): Sequential(
        (0): Conv2dNormActivation(
          (0): Conv2d(96, 576, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(576, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
          (2): ReLU6(inplace=True)
        )
        (1): Conv2dNormActivation(
          (0): Conv2d(576, 576, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), g
roups=576, bias=False)
          (1): BatchNorm2d(576, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
          (2): ReLU6(inplace=True)
        )
        (2): Conv2d(576, 96, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (3): BatchNorm2d(96, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      )
    )
    (13): InvertedResidual(
      (conv): Sequential(
        (0): Conv2dNormActivation(
          (0): Conv2d(96, 576, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(576, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
          (2): ReLU6(inplace=True)
        )
        (1): Conv2dNormActivation(
          (0): Conv2d(576, 576, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), g
roups=576, bias=False)
          (1): BatchNorm2d(576, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
          (2): ReLU6(inplace=True)
        )
        (2): Conv2d(576, 96, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (3): BatchNorm2d(96, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      )
    )
    (14): InvertedResidual(
      (conv): Sequential(
        (0): Conv2dNormActivation(
          (0): Conv2d(96, 576, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(576, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
          (2): ReLU6(inplace=True)
        )
        (1): Conv2dNormActivation(
          (0): Conv2d(576, 576, kernel_size=(3, 3), stride=(2, 2), padding=(1, 1), g
roups=576, bias=False)
          (1): BatchNorm2d(576, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
          (2): ReLU6(inplace=True)
        )
        (2): Conv2d(576, 160, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (3): BatchNorm2d(160, eps=1e-05, momentum=0.1, affine=True, track_running_st
```

```
ats=True)
      )
    )
    (15): InvertedResidual(
      (conv): Sequential(
        (0): Conv2dNormActivation(
          (0): Conv2d(160, 960, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(960, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
          (2): ReLU6(inplace=True)
        )
        (1): Conv2dNormActivation(
          (0): Conv2d(960, 960, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), g
roups=960, bias=False)
          (1): BatchNorm2d(960, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
          (2): ReLU6(inplace=True)
        )
        (2): Conv2d(960, 160, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (3): BatchNorm2d(160, eps=1e-05, momentum=0.1, affine=True, track_running_st
ats=True)
      )
    )
    (16): InvertedResidual(
      (conv): Sequential(
        (0): Conv2dNormActivation(
          (0): Conv2d(160, 960, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(960, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
          (2): ReLU6(inplace=True)
        )
        (1): Conv2dNormActivation(
          (0): Conv2d(960, 960, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), g
roups=960, bias=False)
          (1): BatchNorm2d(960, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
          (2): ReLU6(inplace=True)
        )
        (2): Conv2d(960, 160, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (3): BatchNorm2d(160, eps=1e-05, momentum=0.1, affine=True, track_running_st
ats=True)
      )
    )
    (17): InvertedResidual(
      (conv): Sequential(
        (0): Conv2dNormActivation(
          (0): Conv2d(160, 960, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (1): BatchNorm2d(960, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
          (2): ReLU6(inplace=True)
        )
        (1): Conv2dNormActivation(
          (0): Conv2d(960, 960, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), g
roups=960, bias=False)
          (1): BatchNorm2d(960, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
```

```
          (2): ReLU6(inplace=True)
        )
        (2): Conv2d(960, 320, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (3): BatchNorm2d(320, eps=1e-05, momentum=0.1, affine=True, track_running_st
ats=True)
      )
    )
    (18): Conv2dNormActivation(
      (0): Conv2d(320, 1280, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (1): BatchNorm2d(1280, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
      (2): ReLU6(inplace=True)
    )
  )
  (classifier): Sequential(
    (0): Dropout(p=0.2, inplace=False)
    (1): Linear(in_features=1280, out_features=2, bias=True)
  )
)
```

In [26]:

```
densenet = models.densenet121(weights=models.DenseNet121_Weights.DEFAULT)

# congelar toda la base
freeze_model_base(densenet)

# reemplazar clasificador final
in_features_densenet = densenet.classifier.in_features
densenet.classifier = nn.Linear(in_features_densenet, 2)

densenet = densenet.to(device)
print(densenet)
```

```
Downloading: "https://download.pytorch.org/models/densenet121-a639ec97.pth" to C:\Us
ers\ALEJANDRA/.cache\torch\hub\checkpoints\densenet121-a639ec97.pth
100.0%
```

```
DenseNet(
  (features): Sequential(
    (conv0): Conv2d(3, 64, kernel_size=(7, 7), stride=(2, 2), padding=(3, 3), bias=F
alse)
    (norm0): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_sta
ts=True)
    (relu0): ReLU(inplace=True)
    (pool0): MaxPool2d(kernel_size=3, stride=2, padding=1, dilation=1, ceil_mode=Fal
se)
    (denseblock1): _DenseBlock(
      (denselayer1): _DenseLayer(
        (norm1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running
_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(64, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer2): _DenseLayer(
        (norm1): BatchNorm2d(96, eps=1e-05, momentum=0.1, affine=True, track_running
_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(96, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer3): _DenseLayer(
        (norm1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(128, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer4): _DenseLayer(
        (norm1): BatchNorm2d(160, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(160, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer5): _DenseLayer(
        (norm1): BatchNorm2d(192, eps=1e-05, momentum=0.1, affine=True, track_runnin
```

```
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(192, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer6): _DenseLayer(
        (norm1): BatchNorm2d(224, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(224, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
    )
    (transition1): _Transition(
      (norm): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_s
tats=True)
      (relu): ReLU(inplace=True)
      (conv): Conv2d(256, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (pool): AvgPool2d(kernel_size=2, stride=2, padding=0)
    )
    (denseblock2): _DenseBlock(
      (denselayer1): _DenseLayer(
        (norm1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(128, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer2): _DenseLayer(
        (norm1): BatchNorm2d(160, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(160, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer3): _DenseLayer(
        (norm1): BatchNorm2d(192, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(192, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
```

```
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer4): _DenseLayer(
        (norm1): BatchNorm2d(224, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(224, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer5): _DenseLayer(
        (norm1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(256, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer6): _DenseLayer(
        (norm1): BatchNorm2d(288, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(288, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer7): _DenseLayer(
        (norm1): BatchNorm2d(320, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(320, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer8): _DenseLayer(
        (norm1): BatchNorm2d(352, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(352, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
```

```
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer9): _DenseLayer(
        (norm1): BatchNorm2d(384, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(384, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer10): _DenseLayer(
        (norm1): BatchNorm2d(416, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(416, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer11): _DenseLayer(
        (norm1): BatchNorm2d(448, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(448, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer12): _DenseLayer(
        (norm1): BatchNorm2d(480, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(480, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
    )
    (transition2): _Transition(
      (norm): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_s
tats=True)
      (relu): ReLU(inplace=True)
      (conv): Conv2d(512, 256, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (pool): AvgPool2d(kernel_size=2, stride=2, padding=0)
```

```
    )
    (denseblock3): _DenseBlock(
      (denselayer1): _DenseLayer(
        (norm1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(256, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer2): _DenseLayer(
        (norm1): BatchNorm2d(288, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(288, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer3): _DenseLayer(
        (norm1): BatchNorm2d(320, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(320, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer4): _DenseLayer(
        (norm1): BatchNorm2d(352, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(352, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer5): _DenseLayer(
        (norm1): BatchNorm2d(384, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(384, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
```

```
      )
      (denselayer6): _DenseLayer(
        (norm1): BatchNorm2d(416, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(416, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer7): _DenseLayer(
        (norm1): BatchNorm2d(448, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(448, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer8): _DenseLayer(
        (norm1): BatchNorm2d(480, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(480, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer9): _DenseLayer(
        (norm1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(512, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer10): _DenseLayer(
        (norm1): BatchNorm2d(544, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(544, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
```

```
      (denselayer11): _DenseLayer(
        (norm1): BatchNorm2d(576, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(576, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer12): _DenseLayer(
        (norm1): BatchNorm2d(608, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(608, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer13): _DenseLayer(
        (norm1): BatchNorm2d(640, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(640, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer14): _DenseLayer(
        (norm1): BatchNorm2d(672, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(672, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer15): _DenseLayer(
        (norm1): BatchNorm2d(704, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(704, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer16): _DenseLayer(
```

```
        (norm1): BatchNorm2d(736, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(736, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer17): _DenseLayer(
        (norm1): BatchNorm2d(768, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(768, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer18): _DenseLayer(
        (norm1): BatchNorm2d(800, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(800, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer19): _DenseLayer(
        (norm1): BatchNorm2d(832, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(832, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer20): _DenseLayer(
        (norm1): BatchNorm2d(864, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(864, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer21): _DenseLayer(
        (norm1): BatchNorm2d(896, eps=1e-05, momentum=0.1, affine=True, track_runnin
```

```
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(896, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer22): _DenseLayer(
        (norm1): BatchNorm2d(928, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(928, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer23): _DenseLayer(
        (norm1): BatchNorm2d(960, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(960, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer24): _DenseLayer(
        (norm1): BatchNorm2d(992, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(992, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
    )
    (transition3): _Transition(
      (norm): BatchNorm2d(1024, eps=1e-05, momentum=0.1, affine=True, track_running_
stats=True)
      (relu): ReLU(inplace=True)
      (conv): Conv2d(1024, 512, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (pool): AvgPool2d(kernel_size=2, stride=2, padding=0)
    )
    (denseblock4): _DenseBlock(
      (denselayer1): _DenseLayer(
        (norm1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(512, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
```

```
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer2): _DenseLayer(
        (norm1): BatchNorm2d(544, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(544, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer3): _DenseLayer(
        (norm1): BatchNorm2d(576, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(576, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer4): _DenseLayer(
        (norm1): BatchNorm2d(608, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(608, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer5): _DenseLayer(
        (norm1): BatchNorm2d(640, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(640, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer6): _DenseLayer(
        (norm1): BatchNorm2d(672, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(672, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
```

```
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer7): _DenseLayer(
        (norm1): BatchNorm2d(704, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(704, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer8): _DenseLayer(
        (norm1): BatchNorm2d(736, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(736, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer9): _DenseLayer(
        (norm1): BatchNorm2d(768, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(768, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer10): _DenseLayer(
        (norm1): BatchNorm2d(800, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(800, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer11): _DenseLayer(
        (norm1): BatchNorm2d(832, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(832, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
```

```
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer12): _DenseLayer(
        (norm1): BatchNorm2d(864, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(864, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer13): _DenseLayer(
        (norm1): BatchNorm2d(896, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(896, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer14): _DenseLayer(
        (norm1): BatchNorm2d(928, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(928, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer15): _DenseLayer(
        (norm1): BatchNorm2d(960, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(960, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
      (denselayer16): _DenseLayer(
        (norm1): BatchNorm2d(992, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu1): ReLU(inplace=True)
        (conv1): Conv2d(992, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (norm2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
        (relu2): ReLU(inplace=True)
```

```
        (conv2): Conv2d(128, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1),
bias=False)
      )
    )
    (norm5): BatchNorm2d(1024, eps=1e-05, momentum=0.1, affine=True, track_running_s
tats=True)
  )
  (classifier): Linear(in_features=1024, out_features=2, bias=True)
)
```

In [27]:
```python
def count_trainable_parameters(model):
    total = sum(p.numel() for p in model.parameters())
    trainable = sum(p.numel() for p in model.parameters() if p.requires_grad)
    return total, trainable

total_m, trainable_m = count_trainable_parameters(mobilenet)
total_d, trainable_d = count_trainable_parameters(densenet)

print("MobileNetV2")
print("Total parámetros:", total_m)
print("Entrenables:", trainable_m)
print()

print("DenseNet121")
print("Total parámetros:", total_d)
print("Entrenables:", trainable_d)
```

```
MobileNetV2
Total parámetros: 2226434
Entrenables: 2562

DenseNet121
Total parámetros: 6955906
Entrenables: 2050
```

Evaluación de los modelos

In [29]:
```python
def evaluate_model(model, dataloader, criterion, device):
    model.eval()

    running_loss = 0.0
    all_labels = []
    all_preds = []

    with torch.no_grad():
        for images, labels in dataloader:
            images = images.to(device)
            labels = labels.to(device)

            outputs = model(images)
            loss = criterion(outputs, labels)

            running_loss += loss.item() * images.size(0)

            preds = torch.argmax(outputs, dim=1)
```

```
            all_labels.extend(labels.cpu().numpy())
            all_preds.extend(preds.cpu().numpy())

    epoch_loss = running_loss / len(dataloader.dataset)
    epoch_acc = accuracy_score(all_labels, all_preds)
    epoch_f1 = f1_score(all_labels, all_preds, pos_label=1)
    epoch_recall = recall_score(all_labels, all_preds, pos_label=1)

    return {
        "loss": epoch_loss,
        "accuracy": epoch_acc,
        "f1": epoch_f1,
        "recall": epoch_recall
    }
```

Training loop

In [30]:
```
def train_model(
    model,
    train_loader,
    val_loader,
    criterion,
    optimizer,
    device,
    num_epochs=15,
    patience=3,
    model_name="model"
):
    history = {
        "train_loss": [],
        "train_acc": [],
        "val_loss": [],
        "val_acc": [],
        "val_f1": [],
        "val_recall": []
    }

    best_val_loss = float("inf")
    best_model_wts = copy.deepcopy(model.state_dict())
    epochs_without_improvement = 0

    start_time = time.time()

    for epoch in range(num_epochs):
        model.train()

        running_loss = 0.0
        all_train_labels = []
        all_train_preds = []

        for images, labels in train_loader:
            images = images.to(device)
            labels = labels.to(device)
```

```
            optimizer.zero_grad()

            outputs = model(images)
            loss = criterion(outputs, labels)

            loss.backward()
            optimizer.step()

            running_loss += loss.item() * images.size(0)

            preds = torch.argmax(outputs, dim=1)
            all_train_labels.extend(labels.cpu().numpy())
            all_train_preds.extend(preds.cpu().numpy())

        train_loss = running_loss / len(train_loader.dataset)
        train_acc = accuracy_score(all_train_labels, all_train_preds)

        val_metrics = evaluate_model(model, val_loader, criterion, device)

        history["train_loss"].append(train_loss)
        history["train_acc"].append(train_acc)
        history["val_loss"].append(val_metrics["loss"])
        history["val_acc"].append(val_metrics["accuracy"])
        history["val_f1"].append(val_metrics["f1"])
        history["val_recall"].append(val_metrics["recall"])

        print(f"Epoch [{epoch+1}/{num_epochs}]")
        print(f"Train Loss: {train_loss:.4f} | Train Acc: {train_acc:.4f}")
        print(f"Val Loss:   {val_metrics['loss']:.4f} | Val Acc: {val_metrics['accu
        print(f"Val F1:     {val_metrics['f1']:.4f} | Val Recall: {val_metrics['rec
        print("-" * 50)

        if val_metrics["loss"] < best_val_loss:
            best_val_loss = val_metrics["loss"]
            best_model_wts = copy.deepcopy(model.state_dict())
            epochs_without_improvement = 0

            torch.save(best_model_wts, f"best_{model_name}.pth")
            print(f"Mejor modelo guardado: best_{model_name}.pth")
        else:
            epochs_without_improvement += 1
            print(f"Sin mejora en val_loss por {epochs_without_improvement} época(s

        if epochs_without_improvement >= patience:
            print(f"Early stopping activado en la época {epoch+1}.")
            break

    total_time = time.time() - start_time
    print(f"Entrenamiento completado en {total_time/60:.2f} minutos.")

    model.load_state_dict(best_model_wts)

    return model, history
```

In [31]:
```
class_counts = train_df["label"].value_counts().sort_index()
```

```
class_counts
```

Out[31]:
```
label
0    1108
1    2991
Name: count, dtype: int64
```

In [32]:
```
num_samples = len(train_df)
num_classes = 2

class_weights = num_samples / (num_classes * class_counts)
class_weights = torch.tensor(class_weights.values, dtype=torch.float32).to(device)

print("Class weights:", class_weights)
```

```
Class weights: tensor([1.8497, 0.6852], device='cuda:0')
```

In [35]:
```
criterion_mobilenet = nn.CrossEntropyLoss(weight=class_weights)

optimizer_mobilenet = torch.optim.Adam(
    mobilenet.parameters(),
    lr=1e-3
)

mobilenet, history_mobilenet = train_model(
    model=mobilenet,
    train_loader=train_loader,
    val_loader=val_loader,
    criterion=criterion_mobilenet,
    optimizer=optimizer_mobilenet,
    device=device,
    num_epochs=15,
    patience=3,
    model_name="mobilenetv2"
)
```

```
Epoch [1/15]
Train Loss: 0.2864 | Train Acc: 0.8939
Val Loss:   0.2645 | Val Acc: 0.8599
Val F1:     0.8950 | Val Recall: 0.8175
--------------------------------------------------
Mejor modelo guardado: best_mobilenetv2.pth
Epoch [2/15]
Train Loss: 0.2428 | Train Acc: 0.9088
Val Loss:   0.2284 | Val Acc: 0.8918
Val F1:     0.9210 | Val Recall: 0.8643
--------------------------------------------------
Mejor modelo guardado: best_mobilenetv2.pth
Epoch [3/15]
Train Loss: 0.2166 | Train Acc: 0.9127
Val Loss:   0.2001 | Val Acc: 0.9043
Val F1:     0.9309 | Val Recall: 0.8830
--------------------------------------------------
Mejor modelo guardado: best_mobilenetv2.pth
Epoch [4/15]
Train Loss: 0.2073 | Train Acc: 0.9175
Val Loss:   0.2004 | Val Acc: 0.9009
Val F1:     0.9278 | Val Recall: 0.8721
--------------------------------------------------
Sin mejora en val_loss por 1 época(s).
Epoch [5/15]
Train Loss: 0.2035 | Train Acc: 0.9207
Val Loss:   0.1800 | Val Acc: 0.9248
Val F1:     0.9465 | Val Recall: 0.9111
--------------------------------------------------
Mejor modelo guardado: best_mobilenetv2.pth
Epoch [6/15]
Train Loss: 0.2154 | Train Acc: 0.9197
Val Loss:   0.1882 | Val Acc: 0.9123
Val F1:     0.9370 | Val Recall: 0.8939
--------------------------------------------------
Sin mejora en val_loss por 1 época(s).
Epoch [7/15]
Train Loss: 0.1859 | Train Acc: 0.9288
Val Loss:   0.1864 | Val Acc: 0.9066
Val F1:     0.9322 | Val Recall: 0.8799
--------------------------------------------------
Sin mejora en val_loss por 2 época(s).
Epoch [8/15]
Train Loss: 0.1951 | Train Acc: 0.9263
Val Loss:   0.1690 | Val Acc: 0.9385
Val F1:     0.9569 | Val Recall: 0.9360
--------------------------------------------------
Mejor modelo guardado: best_mobilenetv2.pth
Epoch [9/15]
Train Loss: 0.2007 | Train Acc: 0.9180
Val Loss:   0.1695 | Val Acc: 0.9260
Val F1:     0.9474 | Val Recall: 0.9126
--------------------------------------------------
Sin mejora en val_loss por 1 época(s).
Epoch [10/15]
Train Loss: 0.1839 | Train Acc: 0.9295
```

```
Val Loss:   0.1752 | Val Acc: 0.9191
Val F1:     0.9419 | Val Recall: 0.8970
--------------------------------------------------
Sin mejora en val_loss por 2 época(s).
Epoch [11/15]
Train Loss: 0.1782 | Train Acc: 0.9280
Val Loss:   0.1714 | Val Acc: 0.9226
Val F1:     0.9447 | Val Recall: 0.9064
--------------------------------------------------
Sin mejora en val_loss por 3 época(s).
Early stopping activado en la época 11.
Entrenamiento completado en 15.33 minutos.
```

In [36]:

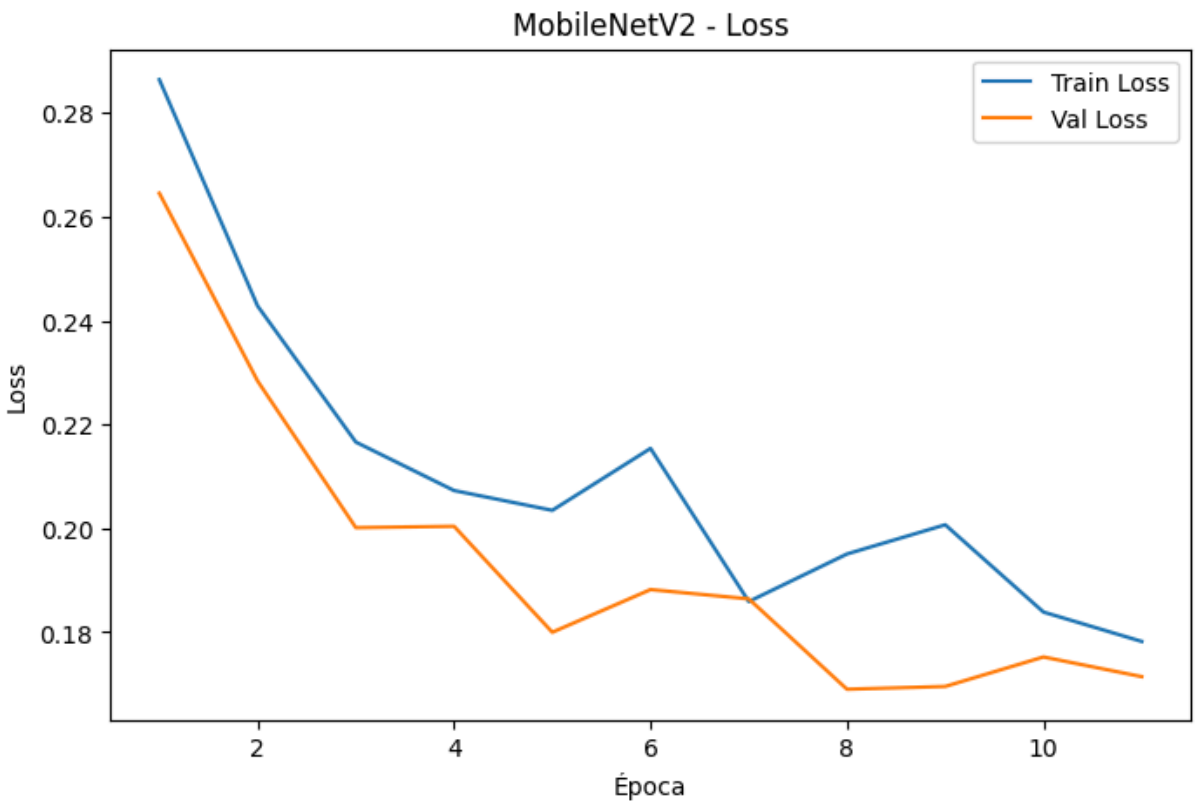
```
criterion_densenet = nn.CrossEntropyLoss(weight=class_weights)

optimizer_densenet = torch.optim.Adam(
    densenet.parameters(),
    lr=1e-3
)

densenet, history_densenet = train_model(
    model=densenet,
    train_loader=train_loader,
    val_loader=val_loader,
    criterion=criterion_densenet,
    optimizer=optimizer_densenet,
    device=device,
    num_epochs=15,
    patience=3,
    model_name="densenet121"
)
```

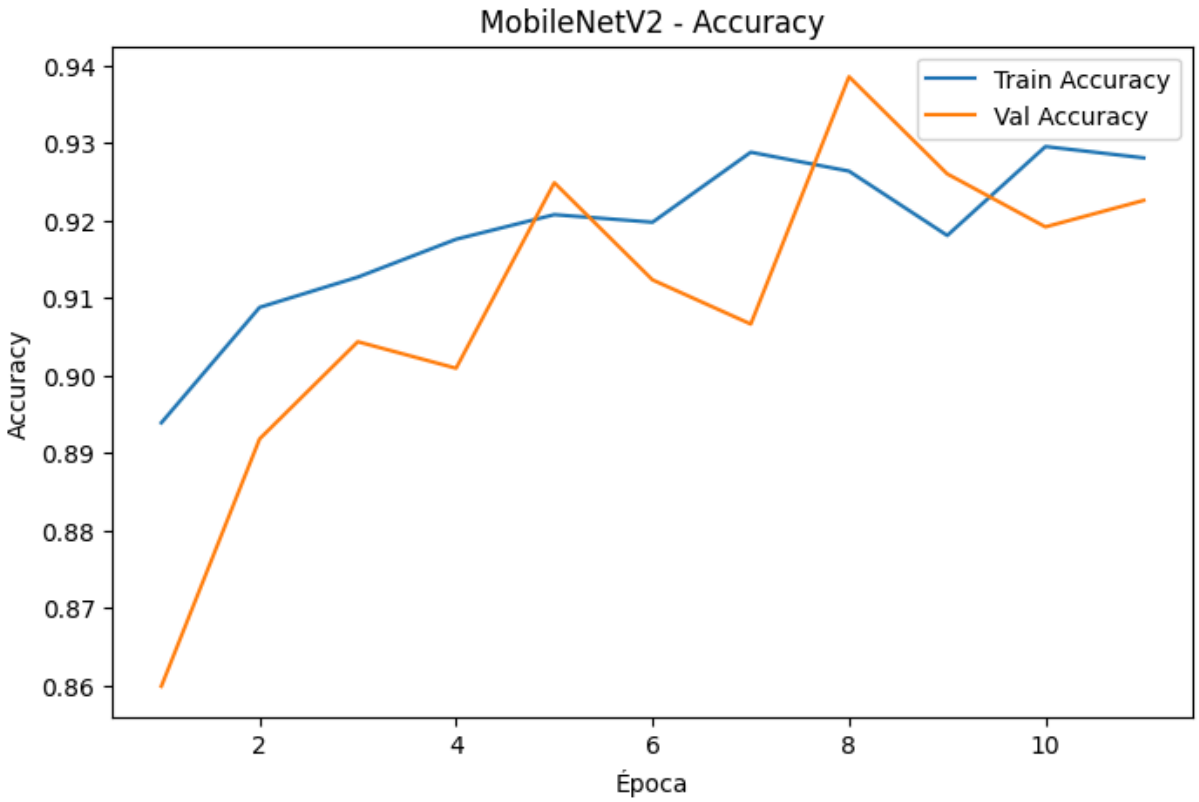
```
Epoch [1/15]
Train Loss: 0.3740 | Train Acc: 0.8607
Val Loss:   0.2797 | Val Acc: 0.8565
Val F1:     0.8919 | Val Recall: 0.8112
--------------------------------------------------
Mejor modelo guardado: best_densenet121.pth
Epoch [2/15]
Train Loss: 0.2426 | Train Acc: 0.9149
Val Loss:   0.2087 | Val Acc: 0.9146
Val F1:     0.9395 | Val Recall: 0.9080
--------------------------------------------------
Mejor modelo guardado: best_densenet121.pth
Epoch [3/15]
Train Loss: 0.2132 | Train Acc: 0.9195
Val Loss:   0.2052 | Val Acc: 0.9021
Val F1:     0.9289 | Val Recall: 0.8768
--------------------------------------------------
Mejor modelo guardado: best_densenet121.pth
Epoch [4/15]
Train Loss: 0.1974 | Train Acc: 0.9249
Val Loss:   0.1971 | Val Acc: 0.9055
Val F1:     0.9315 | Val Recall: 0.8799
--------------------------------------------------
Mejor modelo guardado: best_densenet121.pth
Epoch [5/15]
Train Loss: 0.1891 | Train Acc: 0.9283
Val Loss:   0.1769 | Val Acc: 0.9317
Val F1:     0.9519 | Val Recall: 0.9267
--------------------------------------------------
Mejor modelo guardado: best_densenet121.pth
Epoch [6/15]
Train Loss: 0.1769 | Train Acc: 0.9319
Val Loss:   0.1811 | Val Acc: 0.9112
Val F1:     0.9360 | Val Recall: 0.8892
--------------------------------------------------
Sin mejora en val_loss por 1 época(s).
Epoch [7/15]
Train Loss: 0.1787 | Train Acc: 0.9356
Val Loss:   0.1984 | Val Acc: 0.8952
Val F1:     0.9232 | Val Recall: 0.8627
--------------------------------------------------
Sin mejora en val_loss por 2 época(s).
Epoch [8/15]
Train Loss: 0.1684 | Train Acc: 0.9385
Val Loss:   0.1770 | Val Acc: 0.9146
Val F1:     0.9386 | Val Recall: 0.8939
--------------------------------------------------
Sin mejora en val_loss por 3 época(s).
Early stopping activado en la época 8.
Entrenamiento completado en 12.54 minutos.
```

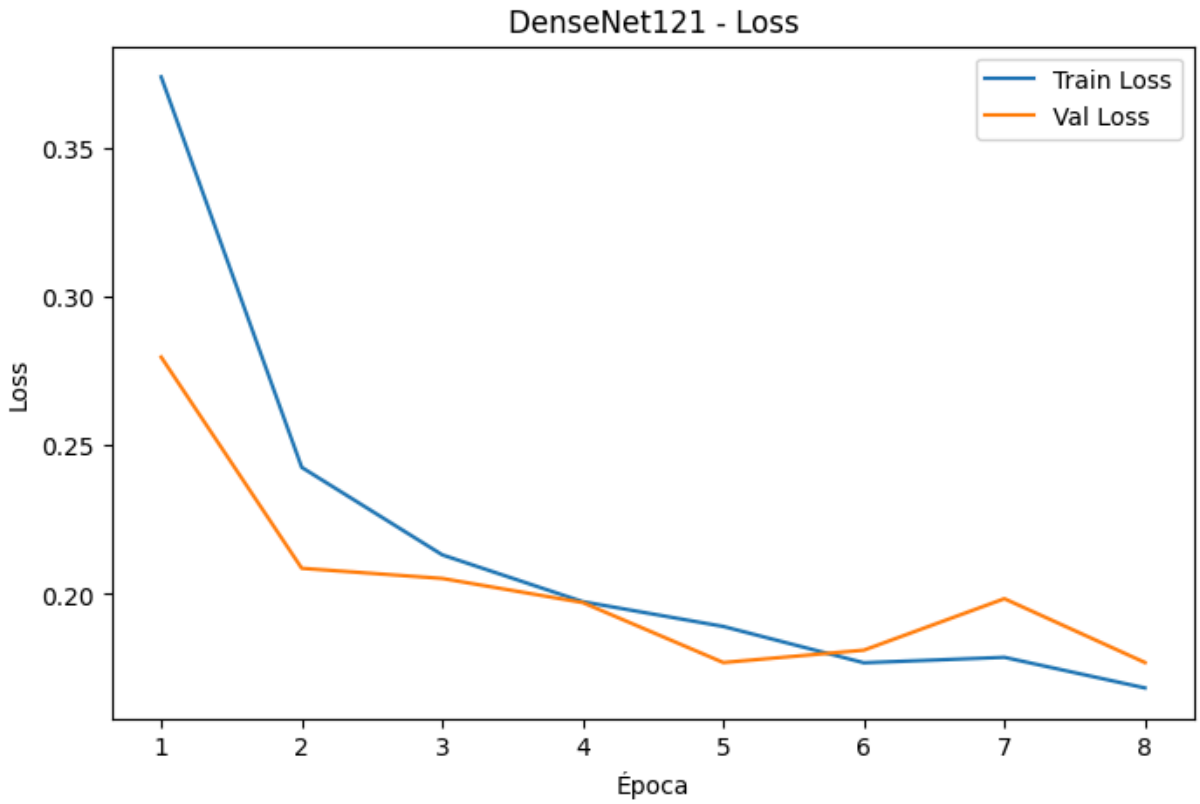
Historial de entrenamiento

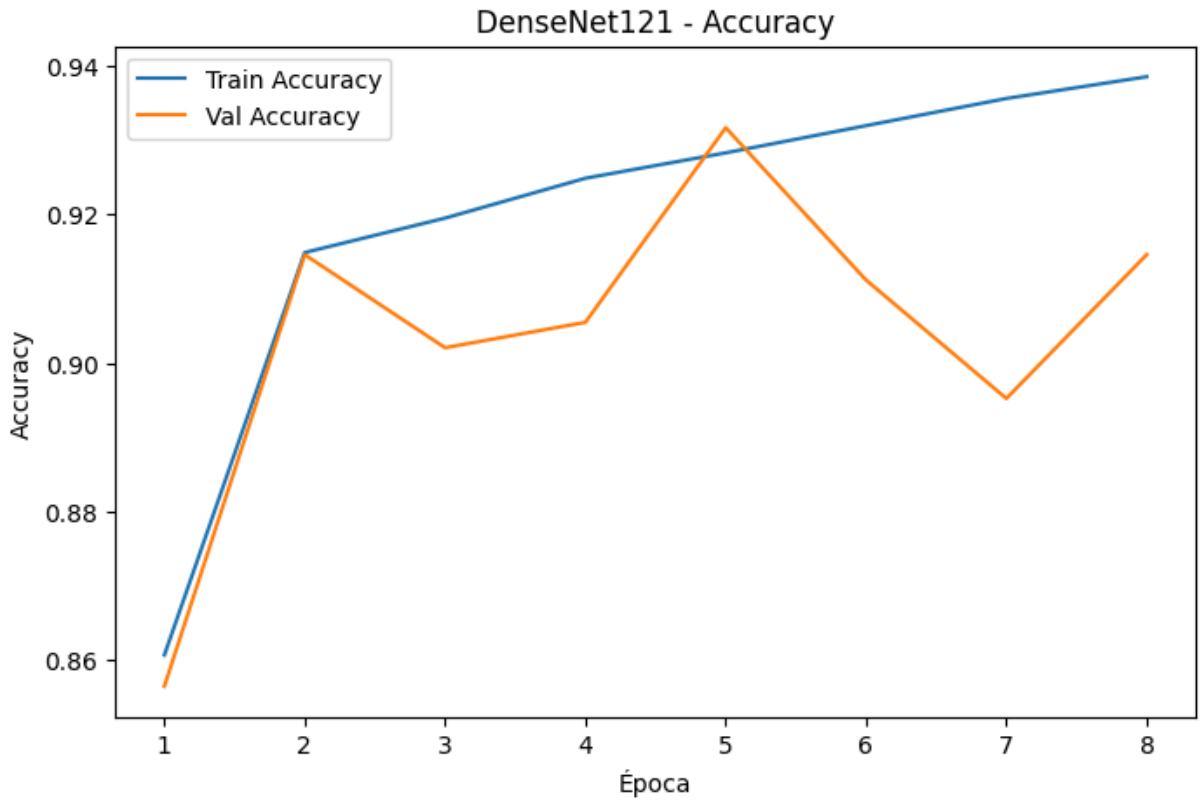
In [ ]:
```python
def plot_history(history, model_name):
    epochs = range(1, len(history["train_loss"]) + 1)
```

```
    plt.figure(figsize=(8, 5))
    plt.plot(epochs, history["train_loss"], label="Train Loss")
    plt.plot(epochs, history["val_loss"], label="Val Loss")
    plt.title(f"{model_name} - Loss")
    plt.xlabel("Época")
    plt.ylabel("Loss")
    plt.legend()
    plt.show()

    plt.figure(figsize=(8, 5))
    plt.plot(epochs, history["train_acc"], label="Train Accuracy")
    plt.plot(epochs, history["val_acc"], label="Val Accuracy")
    plt.title(f"{model_name} - Accuracy")
    plt.xlabel("Época")
    plt.ylabel("Accuracy")
    plt.legend()
    plt.show()

plot_history(history_mobilenet, "MobileNetV2")
plot_history(history_densenet, "DenseNet121")
```




MobileNetV2 - Accuracy
Train Accuracy
Val Accuracy
Accuracy
0.94
0.93
0.92
0.91
0.90
0.89
0.88
0.87
0.86
2
4
6
8
10
Época


DenseNet121 - Loss
Train Loss
Val Loss
Loss
0.35
0.30
0.25
0.20
1
2
3
4
5
6
7
8
Época



In [40]:
```
def test_model(model, dataloader, device):
    model.eval()

    all_labels = []
    all_preds = []

    with torch.no_grad():
        for images, labels in dataloader:
            images = images.to(device)
            labels = labels.to(device)

            outputs = model(images)
            preds = torch.argmax(outputs, dim=1)

            all_labels.extend(labels.cpu().numpy())
            all_preds.extend(preds.cpu().numpy())

    accuracy = accuracy_score(all_labels, all_preds)
    f1_pneumonia = f1_score(all_labels, all_preds, pos_label=1)
    recall_pneumonia = recall_score(all_labels, all_preds, pos_label=1)

    return {
        "accuracy": accuracy,
        "f1_pneumonia": f1_pneumonia,
        "recall_pneumonia": recall_pneumonia
    }
```

In [41]:
```
mobilenet_test_metrics = test_model(mobilenet, test_loader, device)
densenet_test_metrics = test_model(densenet, test_loader, device)
```

```
print("=== MobileNetV2 - Test ===")
for k, v in mobilenet_test_metrics.items():
    print(f"{k}: {v:.4f}")

print("\n=== DenseNet121 - Test ===")
for k, v in densenet_test_metrics.items():
    print(f"{k}: {v:.4f}")
```

```
=== MobileNetV2 - Test ===
accuracy: 0.9158
f1_pneumonia: 0.9410
recall_pneumonia: 0.9204

=== DenseNet121 - Test ===
accuracy: 0.9261
f1_pneumonia: 0.9481
recall_pneumonia: 0.9267
```

In [42]:
```
def get_model_size_mb(model_path):
    size_bytes = os.path.getsize(model_path)
    size_mb = size_bytes / (1024 * 1024)
    return size_mb
```

In [43]:
```
mobilenet_size_mb = get_model_size_mb("best_mobilenetv2.pth")
densenet_size_mb = get_model_size_mb("best_densenet121.pth")

print(f"MobileNetV2 size: {mobilenet_size_mb:.2f} MB")
print(f"DenseNet121 size: {densenet_size_mb:.2f} MB")
```

```
MobileNetV2 size: 8.73 MB
DenseNet121 size: 27.12 MB
```

In [54]:
```
def measure_inference_time(model, dataset, device, num_images=100):
    model.eval()

    num_images = min(num_images, len(dataset))
    times = []

    with torch.no_grad():
        for i in range(num_images):
            image, _ = dataset[i]
            image = image.unsqueeze(0).to(device)

            if device.type == "cuda":
                torch.cuda.synchronize()

            start_time = time.perf_counter()
            _ = model(image)

            if device.type == "cuda":
                torch.cuda.synchronize()

            end_time = time.perf_counter()

            elapsed_ms = (end_time - start_time) * 1000
            times.append(elapsed_ms)
```

```
    return np.mean(times)
```

In [55]:
```
mobilenet_inference_ms = measure_inference_time(mobilenet, test_dataset, device, nu
densenet_inference_ms = measure_inference_time(densenet, test_dataset, device, num_

print(f"MobileNetV2 avg inference time (100 images): {mobilenet_inference_ms:.2f} m
print(f"DenseNet121 avg inference time (100 images): {densenet_inference_ms:.2f} ms
```

```
MobileNetV2 avg inference time (100 images): 8.37 ms/image
DenseNet121 avg inference time (100 images): 26.69 ms/image
```

In [56]:
```
results_df = pd.DataFrame([
    {
        "Model": "MobileNetV2",
        "Accuracy": mobilenet_test_metrics["accuracy"],
        "F1_Pneumonia": mobilenet_test_metrics["f1_pneumonia"],
        "Recall_Pneumonia": mobilenet_test_metrics["recall_pneumonia"],
        "Size_MB": mobilenet_size_mb,
        "Inference_ms_per_image": mobilenet_inference_ms
    },
    {
        "Model": "DenseNet121",
        "Accuracy": densenet_test_metrics["accuracy"],
        "F1_Pneumonia": densenet_test_metrics["f1_pneumonia"],
        "Recall_Pneumonia": densenet_test_metrics["recall_pneumonia"],
        "Size_MB": densenet_size_mb,
        "Inference_ms_per_image": densenet_inference_ms
    }
])
```

In [57]:
```
results_df_rounded = results_df.copy()
results_df_rounded["Accuracy"] = results_df_rounded["Accuracy"].round(4)
results_df_rounded["F1_Pneumonia"] = results_df_rounded["F1_Pneumonia"].round(4)
results_df_rounded["Recall_Pneumonia"] = results_df_rounded["Recall_Pneumonia"].rou
results_df_rounded["Size_MB"] = results_df_rounded["Size_MB"].round(2)
results_df_rounded["Inference_ms_per_image"] = results_df_rounded["Inference_ms_per

results_df_rounded
```

Out[57]:

	Model	Accuracy	F1_Pneumonia	Recall_Pneumonia	Size_MB	Inference_ms_per_im
0	MobileNetV2	0.9158	0.9410	0.9204	8.73	
1	DenseNet121	0.9261	0.9481	0.9267	27.12	2

In [58]:
```
results_df_rounded["F1_percent"] = results_df["F1_Pneumonia"] * 100
results_df_rounded["MB_per_F1_point"] = results_df["Size_MB"] / (results_df["F1_Pne

results_df_rounded[["Model", "Size_MB", "F1_percent", "MB_per_F1_point"]]
```

Out[58]:

	Model	Size_MB	F1_percent	MB_per_F1_point
0	MobileNetV2	8.73	94.098884	0.092767
1	DenseNet121	27.12	94.812450	0.286017

In [59]:

```
results_df_rounded["Images_per_second"] = 1000 / results_df["Inference_ms_per_image
results_df_rounded["Images_per_hour"] = results_df_rounded["Images_per_second"] * 3

results_df_rounded[["Model", "Inference_ms_per_image", "Images_per_second", "Images
```

Out[59]:

	Model	Inference_ms_per_image	Images_per_second	Images_per_hour
0	MobileNetV2	8.37	119.497627	430191.457675
1	DenseNet121	26.69	37.465523	134875.883835

Prompt utilizado

Para la implementación inicial del pipeline, se utilizó un prompt básico de apoyo para estructurar el flujo general del proyecto en PyTorch. El prompt utilizado fue:

"*How can I build a basic image classification pipeline in PyTorch including dataset loading, train/validation split, data augmentation, and model training?*"

Este prompt permitió obtener un esqueleto general del proceso, incluyendo la organización del código en etapas (carga de datos, transformaciones, modelo y entrenamiento). Sin embargo, la solución final no se basó directamente en la salida generada, sino que fue adaptada y refinada manualmente. SEn particular, se realizaron decisiones específicas del problema, como la unificación del dataset original, la implementación de una división estratificada 70/15/15, el diseño de un pipeline de data augmentation adecuado para imágenes médicas y la selección y modificación de modelos preentrenados.